

Longitudinal Data Analysis with R Statistical Software

Day 5: Joint models for longitudinal and survival data

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Day 4 review

Overview

Motivation and examples

Longitudinal data analysis

Survival data analysis

Joint regression models

Example 1

Longitudinal changes in peripheral monocytes (Yoshida et al., 2019)

- **Adult Health Study**

- ▶ Subset of Life Span Study of atomic bomb survivors
- ▶ Biennial clinic examinations since 1958
- ▶ Detailed questionnaire and laboratory data

- DS02R1 radiation doses estimated from dosimetry system

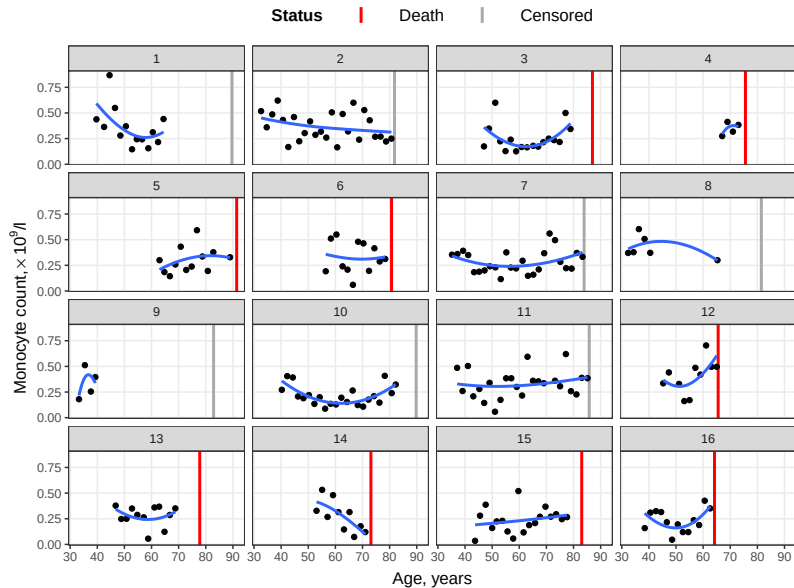
- **Outcomes of interest**

- ▶ Monocyte count (longitudinal) as a measure of inflammation
- ▶ Time to death due to any cause (1958–2010)

- **Research questions**

- ▶ What is the association between radiation and monocyte counts?
- ▶ What is the association between monocyte counts and mortality?
- ▶ Others?

AHS data

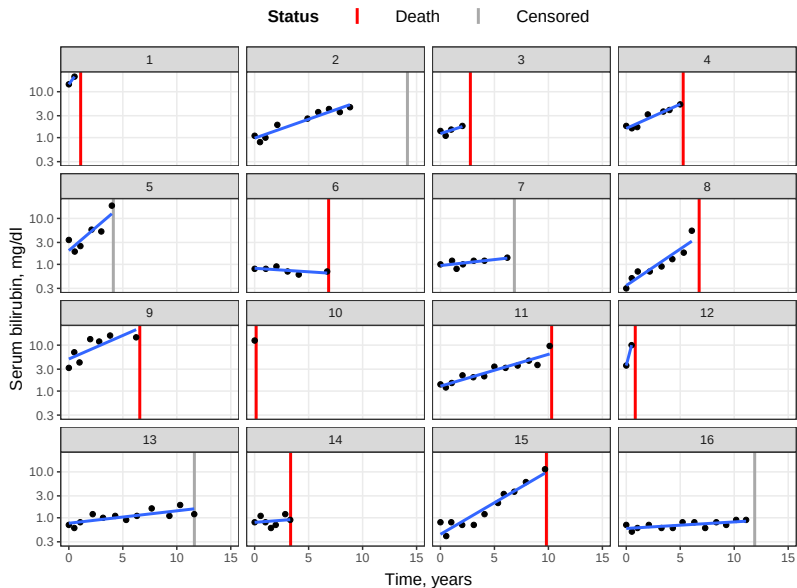


Example 2

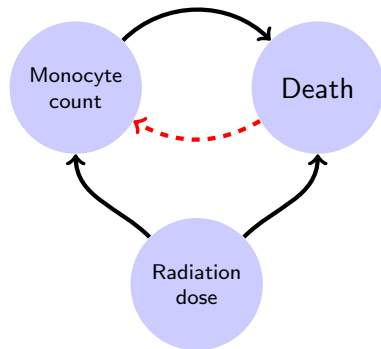
Mayo Clinic trial in primary biliary cirrhosis (Murtaugh et al., 1994)

- **Primary biliary cirrhosis**
 - ▶ Chronic and fatal but rare liver disease
 - ▶ Inflammatory destruction of small bile ducts within the liver
 - ▶ Patients referred to Mayo Clinic, 1974–1984
- 158 patients randomized to treatment with D-penicillamine;
154 randomized to placebo
- **Outcomes of interest**
 - ▶ Serum bilirubin levels (longitudinal) as a measure of liver function
 - ▶ Time to death and/or time to liver transplantation
- **Research questions**
 - ▶ What is the association between treatment and serum bilirubin levels?
 - ▶ What is the association between serum bilirubin levels and mortality?
 - ▶ Others?

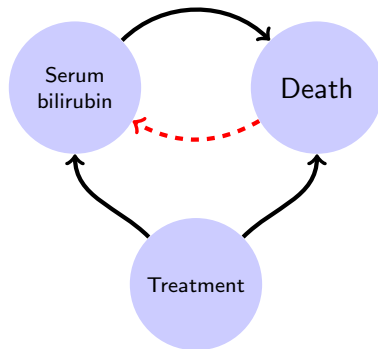
PBC data



Analysis framework



(a) AHS data



(b) PBC data

- Treatment can be associated with both bilirubin and mortality
- Bilirubin levels (and trends) can be associated with mortality
- Occurrence of death precludes observation of bilirubin levels

Analysis choices

Analysis methods should be selected based on the scientific question

- Focus on the association between treatment and bilirubin levels
 - ▶ Standard **longitudinal data analysis** (separate analysis)
 - ▶ Formulate a regression model for repeated measures of bilirubin level
 - ▶ Ignores the impact of death on our ability to collect data (—)
 - Focus on the association between bilirubin levels and mortality risk
 - ▶ Standard **survival data analysis** (separate analysis)
 - ▶ Formulate a Cox regression model for time to death or transplantation
 - ▶ Potentially adjust for risk differences due to treatment; mediation
 - ▶ Treats bilirubin levels as fixed, but these are measured with error (—)
- (—)s motivate application of **joint regression models** (joint analysis)

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Analysis approaches

Must account for **correlation** due to repeated measurements over time

- Failure to account for correlation $\Rightarrow$ incorrect standard estimates, resulting in incorrect confidence intervals and hypothesis tests
- **Approaches:** Include all observed data in a regression model for the mean response and account for longitudinal correlation
 - ▶ **Generalized estimating equations (GEE):** A marginal model for the mean response and a model for longitudinal correlation

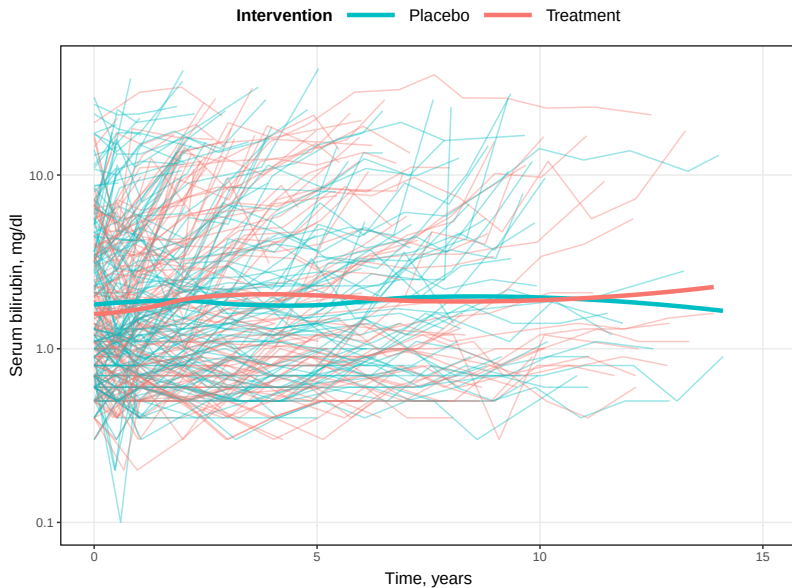
$$g(E[Y_{ij} | x_{ij}]) = x_{ij}\beta \quad \text{and} \quad \text{Corr}[Y_{ij}, Y_{ij'}] = \rho(\alpha), j \neq j'$$

- ▶ **Generalized linear mixed-effects models (GLMM):** A conditional model for the mean response given subject-specific random effects, which induce a (possibly hierarchical) correlation structure

$$g(E[Y_{ij} | x_{ij}, b_i]) = x_{ij}\beta + z_{ij}b_i \quad \text{with} \quad b_i \sim N(0, D)$$

NB: Differences in interpretation of β between GEE and GLMM

Repeated measures



Missing data

- Missing values arise in longitudinal studies whenever the intended serial observations collected on a subject over time are incomplete
 - ▶ Collect fewer data than planned $\Rightarrow$ decreased efficiency (power)
 - ▶ Missingness can depend on outcome values $\Rightarrow$ potential bias
- Important to distinguish between missing data and unbalanced data, although missing data necessarily result in unbalanced data
- Missing data require consideration of the factors that influence the missingness of intended observations
- Also important to distinguish between intermittent missing values (non-monotone) and dropouts in which all observations are missing after subjects are lost to follow-up (monotone)

Pattern	t_1	t_2	t_3	t_4	t_5
Monotone	3.8	3.1	2.0		
Non-monotone	4.1		3.8		

Mechanisms

Partition the complete set of intended observations into the observed and missing data; what factors influence missingness of intended observations?

- **Missing completely at random (MCAR)**

Missingness does not depend on **either** the observed or missing data

- ▶ Example: Administrative censoring at a fixed calendar time
- ▶ Mixed-effects models are valid

- **Missing at random (MAR)**

Missingness depends **only** on the observed data

- ▶ Example: Individuals with no current weight loss in a weight-loss study
- ▶ Mixed-effects models are valid (with additional assumptions)

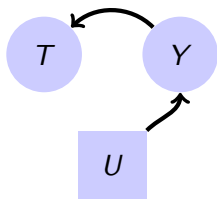
- **Missing not at random (MNAR)**

Missingness depends on **both** the observed and missing data

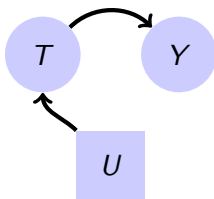
- ▶ Example: Subjects in a prospective study based on disease prognosis
- ▶ Mixed-effects models are not valid

(Rubin, 1976; Ibrahim and Molenberghs, 2009)

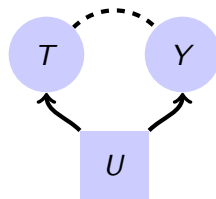
Models under MNAR



(a) Selection



(b) Pattern-mixture



(c) Shared-parameter

- (a) Subject-specific random effects or latent characteristics influence the outcome, which subsequently determines the propensity to drop out
- (b) Subject-specific characteristics initially determine propensity to drop out, with consequential variation in the outcome between drop-out cohorts
- (c) Outcome and drop-out processes jointly respond to unobserved subject-specific characteristics, observation of which would convert the model into one in which the outcome and drop-out time are independent

★ Joint regression models and shared-parameter models are analogous

Summary

- Mixed-effects models assume that each subject has a regression model characterized by subject-specific parameters; a combination of
 - ▶ Fixed-effects parameters common to all individuals in the population
 - ▶ Random-effects parameters unique to each individual subject
- Estimation and inference can focus both on average outcome levels and trends, and on heterogeneity across subjects in levels and trends
- Subject-specific random effects induce a correlation structure
- Must consider the potential for bias due to missing data (drop-out)
 - ▶ Ignores the impact of death on our ability to collect data (—)
- Further reading: Verbeke and Molenberghs (2000)

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Censoring

Analysis methods must account for **censoring** of event-time outcomes

- **Types of censoring**

- ▶ Location of true event time w.r.t. censoring time: right, left, interval
- ▶ Probabilistic relationship between true event time and censoring time: informative, non-informative (similar to MNAR and MAR)

- **Implications**

- ▶ Standard approaches (e.g., t test, linear regression) cannot be used
- ▶ Inference can be sensitive to the assumed distribution of event times

★ **Our focus:** Non-informative right censoring

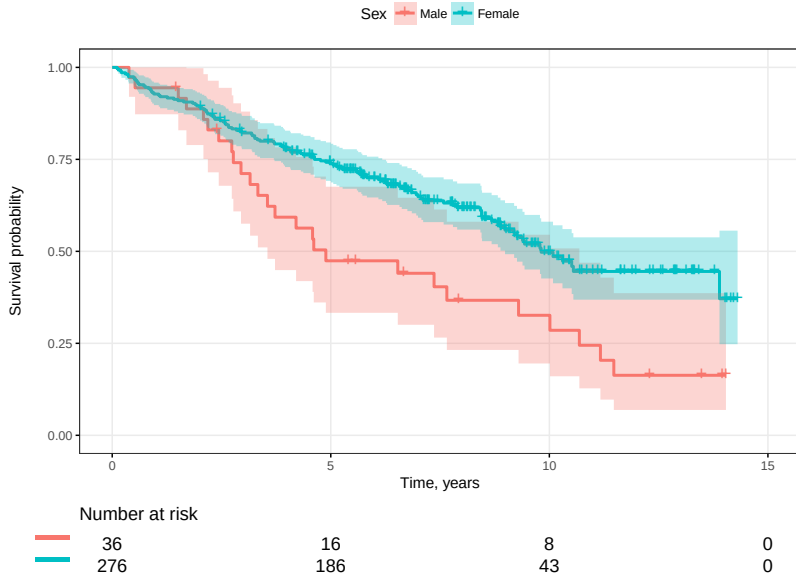
- Event of interest is not fully observed for all subjects

- ▶ Do not experience the event before the end of the study period
- ▶ Are (randomly) lost to follow-up during the study period

⇒ event time (or, survival time) is not known for these subjects

- But, subjects contribute time at risk up until their censoring time

Censoring



Set-up

- **Notation**

- ▶ True event time T_i
- ▶ Censoring time C_i

- **Observed data**

- ▶ Observed event time $T_i^* = \min(T_i, C_i)$
- ▶ Event indicator $\delta_i = 1$ if event observed, $\delta_i = 0$ if censored

- **Objective:** Generate inference for T_i using $\{T_i^*, \delta_i\}$

Hazard functions

- **Probability density function** for the true event time T : $f_T(t)$
- **Cumulative distribution function**: Probability of failure by time t

$$F_T(t) = P[T \leq t] = \int_0^t f_T(s) ds$$

- **Survival function**: Probability of survival beyond time t

$$S_T(t) = P[T > t] = 1 - F_T(t) = \int_t^\infty f_T(s) ds$$

- **Hazard function**: Probability of failure in a very small time period between time t and $t + \Delta t$ given survival up until time t

$$h_T(t) = \frac{f_T(t)}{S_T(t)} = \lim_{\Delta t \rightarrow 0} \frac{P[t \leq T < t + \Delta t \mid T \geq t]}{\Delta t} = -\frac{\partial}{\partial t} \log S_T(t)$$

as a measure of risk; larger hazard $\Rightarrow$ larger risk of failure

Hazard models

- **Relative risk model:** **Multiplicative** effect of covariates on hazard

$$\begin{aligned}h_i(t \mid w_i) &= h_0(t) \exp\{w_i \gamma\} \\ \log h_i(t \mid w_i) &= \log h_0(t) + \gamma_1 w_{i1} + \gamma_2 w_{i2} + \cdots + \gamma_p w_{ip}\end{aligned}$$

with

- ▶ $h_i(t \mid w_i)$: hazard of an event for subject i at time t
 - ▶ $h_0(t)$: common baseline hazard function
 - ▶ $w_i = \{w_{i1}, w_{i2}, \dots, w_{ip}\}$: time-independent covariates for subject i
 - ▶ $\gamma = \{\gamma_1, \gamma_2, \dots, \gamma_p\}^T$: regression parameters of interest
- **Excess relative risk model:** **Additive** effect of covariates on hazard

$$\begin{aligned}h_i(t \mid w_i) &= h_0(t) \{1 + w_i \gamma\} \\ \log h_i(t \mid w_i) &= \log h_0(t) + \log\{1 + \gamma_1 w_{i1} + \gamma_2 w_{i2} + \cdots + \gamma_p w_{ip}\}\end{aligned}$$

which is a standard model for estimating radiation effects

Hazard models

- **Relative risk model:** Multiplicative effect of covariates on hazard

$$h(t \mid w_1) = h_0(t) \exp\{\gamma_1 w_1\}$$

$$h(t \mid w_1 = w) = h_0(t) \exp\{\gamma_1 \times w\}$$

$$h(t \mid w_1 = w + 1) = h_0(t) \exp\{\gamma_1 \times (w + 1)\}$$

$$\frac{h(t \mid w_1 = w + 1)}{h(t \mid w_1 = w)} = \exp(\gamma_1)$$

so that $\exp(\gamma_1)$ is a **hazard ratio** quantifying the impact of a one-unit increase in w_1 on the hazard of an event

- ▶ Does not depend on time t

Estimation and inference

Cox regression model: Unspecified baseline hazard function

- Semi-parametric: No assumption for distribution of event times
- Assumes **proportional hazards** across covariate levels
- Estimates and standard errors obtained from (log) partial likelihood

$$\ell(\gamma) = \sum_{i=1}^n \delta_i \left[w_i \gamma - \log \left\{ \sum_{T_{i'} \geq T_i} \exp(w_{i'} \gamma) \right\} \right]$$

which is a measure of how well the model orders (ranks) the subjects w.r.t. their survival time

- ▶ 'Partial' likelihood because $h_0(t)$ is not involved in estimation

(Cox, 1972)

Time-dependent covariates

- Often interested in the association of a **time-dependent covariate**
 - ▶ AHS: Association between monocyte counts and mortality
 - ▶ PBC: Association between serum bilirubin levels and mortality
- Standard Cox model is appropriate for time-independent covariates
 - ▶ AHS: City, sex, birth cohort, radiation dose
 - ▶ PBC: Treatment, sex, baseline age, baseline serum bilirubin
- Cox model can be extended for a certain type of covariate
 - ▶ External or **exogenous** time-dependent covariate (+)
 - ▶ Internal or **endogenous** time-dependent covariate (−)

Time-dependent covariates

- **Exogenous:** Future path of the covariate up until any time $t > t'$ is not affected by the occurrence of an event at time t'

$$P[\mathcal{Y}_i(t) \mid \mathcal{Y}_i(t'), T_i \geq t'] = P[\mathcal{Y}_i(t) \mid \mathcal{Y}_i(t'), T_i = t']$$

with $0 < t' \leq t$ and $\mathcal{Y}_i(t) = \{y_i(s), 0 \leq s < t\}$ denotes the history

- ▶ Allows the covariate to be associated with the failure rate
 - ▶ But its future values are the same whether a failure occurs or not
 - ▶ Examples: Season of the year, treatment regimen, air pollution level
- **Endogenous:** Not exogenous
 - ▶ Typically arise as time-dependent measurements on study subjects
 - ▶ Subject must survive in order for the covariate to exist
 - ▶ Examples: Monocyte count (AHS), serum bilirubin level (PBC)

Extended Cox model

- Model formulated using counting processes (Andersen and Gill, 1982)

$$h_i(t \mid \mathcal{Y}_i(t), w_i) = h_0(t)R_i(t) \exp\{w_i\gamma + \alpha y_i(t)\}$$

with

- ▶ $\{N_i(t), R_i(t)\}$: event process for subject i
 - ▶ $N_i(t)$: number of events for subject i by time t
 - ▶ $R_i(t)$: at-risk indicator for subject i at time t
 - ▶ $h_i(t)$: intensity process for $N_i(t)$
 - ▶ $h_0(t)$: common baseline intensity function
 - ▶ $w_i = \{w_{i1}, w_{i2}, \dots, w_{ip}\}$: time-independent covariates for subject i
 - ▶ $y_i(t)$: time-dependent covariate for subject i at time t
 - ▶ $\alpha, \gamma = \{\gamma_1, \gamma_2, \dots, \gamma_p\}^T$: regression parameters of interest
- $\exp(\alpha)$ represents the relative increase in risk of an event at time t that results from a simultaneous one-unit increase in $y_i(t)$

Extended Cox model

- Estimates and standard errors obtained from (log) partial likelihood

$$\ell(\alpha, \gamma) = \sum_{i=1}^n \int_0^{\infty} \left[R_i(t) \exp\{w_i \gamma + \alpha y_i(t)\} - \log \sum_{i'} R_{i'}(t) \exp\{w_{i'} \gamma + \alpha y_{i'}(t)\} \right] dN_i(t)$$

- Valid estimation and inference requires several assumptions
 - ▶ Existence of covariate is not related to failure status (exogenous)
 - ▶ Covariate levels are fixed and known in between measurement times; French and Heagerty (2009): $y_i(t)$ exists only at measurement times
 - ▶ Covariate values are free of measurement error

★ Cox regression analysis of an endogenous time-dependent covariate can provide spurious results

Summary

- Analysis methods must account for censoring of event-time outcomes
 - ▶ Do not experience the event before the end of the study period
 - ▶ Are (randomly) lost to follow-up during the study period
- Focus on multiplicative effect of covariates on hazard (risk)
- Semi-parametric Cox regression model under proportional hazards
- Extended Cox model for exogenous time-dependent covariates
 - ▶ Existence of covariate is not related to failure status
 - ▶ Covariate levels are constant in between measurement times
 - ▶ Covariate values are free of measurement error
- Further reading
 - ▶ Kalbfleisch and Prentice (2002)
 - ▶ Cologne et al. (2012): Choice of primary time scale

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Joint regression models

Review

Mixed-effects models for longitudinal data analysis (separate analysis)

- Estimate population-level and subject-specific levels and trends
- Quantify heterogeneity across subjects in outcome levels and trends
- Subject-specific random effects induce a correlation structure
- Must consider the potential for bias due to missing data (—)

Cox regression models for survival data analysis (separate analysis)

- Focus on multiplicative effect of covariates on event hazard
- Account for non-informative right censoring of event-time outcomes
- Appropriate only for exogenous time-dependent covariates (—)
- Assume that covariate values are free of measurement error (—)

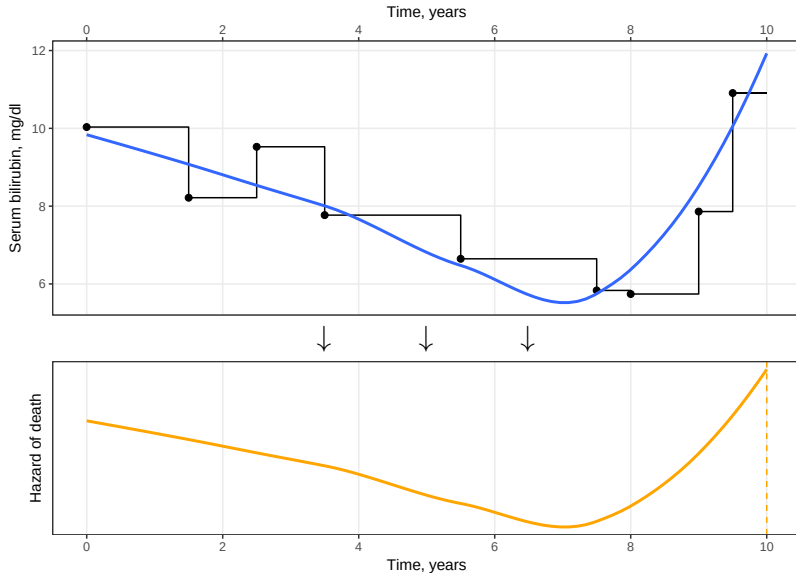
★ Motivates development of **joint regression models** (joint analysis)

Joint models

Longitudinal model + Survival model = Joint model

1. Describe **evolution** of marker levels over time for each subject
 - ▶ Subject-specific random effects
 - ▶ Flexible adjustment for temporal trends
 - ▶ Incorporates random error term for measurement error
2. Associate the subject-specific evolutions with **event hazard**
 - ▶ Allow censoring of event-time outcome
 - ▶ Accommodate marker as an endogenous time-dependent covariate
 - ▶ Marker is not assumed constant in between measurement times
3. Specify a **dependence structure** between the two models

Joint models



Set-up

Step 1: Describe evolution of marker levels over time for each subject

$$\begin{aligned}y_i(t) &= x_i(t)\beta + z_i(t)b_i + \epsilon_i(t) \\ &= m_i(t) + \epsilon_i(t)\end{aligned}$$

using a standard **linear mixed-effects model** with

- True marker value $m_i(t) = x_i(t)\beta + z_i(t)b_i$
- Fixed-effects parameters β
- Random-effects parameters $b_i \sim N(0, D)$
- Measurement error $\epsilon_i(t) \sim N(0, \sigma^2)$

NB: Covariate vectors and error terms are functions of time t

Set-up

Step 2: Associate the subject-specific evolutions with event hazard

$$\begin{aligned}h_i(t \mid \mathcal{M}_i(t), w_i) &= \lim_{\Delta t \rightarrow 0} P[t \leq T < t + \Delta t \mid T \geq t, \mathcal{M}_i(t), w_i] / \Delta t \\&= h_0(t) \exp\{w_i \gamma + \alpha m_i(t)\}\end{aligned}$$

using a standard **relative risk model** with

- Marker history $\mathcal{M}_i(t) = \{m_i(s), 0 \leq s < t\}$
- Common baseline hazard function $h_0(t)$
- Parameters γ for time-independent covariates w_i
- $\exp(\alpha)$ represents the relative increase in risk of an event at time t that results from a simultaneous one-unit increase in $m_i(t)$

Set-up

Step 3: Specify a dependence structure between the two models

1. Define a **joint distribution** for markers y_i and event times $\{T_i^*, \delta_i\}$

$$f(y_i, T_i^*, \delta_i) = \int f_Y(y_i | b_i) h(T_i^* | b_i)^{\delta_i} S(T_i^* | b_i) f_b(b_i) db_i$$

with

- ▶ f_Y : density function for markers y_i
- ▶ f_b : density function for random effects b_i
- ▶ $h(t)$: hazard function for event times T_i^*
- ▶ $S(t)$: survival function for event times T_i^*
- ▶ δ_i : event indicator (1, event; 0, censored)
- ▶ b_i : random effects for inter-dependencies

(Tsiatis and Davidian, 2004)

Set-up

Step 3: Specify a dependence structure between the two models

2. Assume **full conditional independence** given random effects

- ▶ Repeated measurements (longitudinal) are mutually independent

$$y_{ij} \perp\!\!\!\perp y_{ij'} \mid b_i, j \neq j'$$

- ★ Random effects explain the correlation among repeated measures
- ▶ Longitudinal outcome and event-time outcome are independent

$$y_i \perp\!\!\!\perp \{T_i^*, \delta_i\} \mid b_i \forall i$$

- ★ Random effects explain the association between the longitudinal and event-time outcomes (recall shared-parameter models)
- ★ Random effects explain all the inter-dependencies in the data

NB: Conditional independence is difficult to evaluate using observed data

Assumptions

- Censoring and observation-time processes are non-informative; study withdrawal or appearance at study visits for data collection
 - ▶ Can depend on observed history (covariates and previous responses)
 - ▶ Cannot depend on unobserved characteristics associated with prognosis

Tan et al. (2014): Dependence on an unobserved latent variable

- Survival function depends on the complete history of the marker

$$\begin{aligned} S_i(t \mid \mathcal{M}_i(t), w_i) &= P[T_i > t \mid \mathcal{M}_i(t), w_i] \\ &= \exp \left(\int_0^t h_0(s) \exp\{w_i \gamma + \alpha m_i(s)\} ds \right) \end{aligned}$$

so that development of the mixed-effects model should consider

- ▶ Correct specification for subject-specific random effects
- ▶ Flexible adjustment for temporal trends (polynomials, splines)
- ▶ Time interactions with time-independent covariates

Assumptions

Baseline hazard function $h_0(t)$

- Unspecified baseline hazard function in standard Cox regression
 - ▶ Avoid an assumption for the distribution of event times (+)
 - ▶ In joint model can result in under-estimation of standard errors (–)
- Parametric but with different levels of flexibility
 - ▶ **Fully parametric:** Risk function for a parametric distribution
 - ★ Exponential: $h_0(t) = \lambda$
 - ★ Weibull: $h_0(t) = \lambda at^{a-1}$
 - ▶ **Somewhat parametric:** Parametric but flexible risk function
 - ★ Piecewise constant step function
 - ★ Regression splines

but avoid an over-specified model with too many parameters;
between 1/20 and 1/10 of the total number of events

Estimation and inference

Given the set-up and assumptions. . .

- Likelihood **estimation** for $\{\alpha, \beta, \gamma, \sigma, D\}$ given data $\{y_i, T_i^*, \delta_i\}$

$$\ell(\alpha, \beta, \gamma, \sigma, D) = \sum_{i=1}^n \log \int f_Y(y_i | b_i) h(T_i^* | b_i)^{\delta_i} S(T_i^* | b_i) f_b(b_i) db_i$$

with densities f_Y and f_b , hazard $h(t)$ and survival $S(t)$ functions

- ▶ Integration requires numerical approximation
 - ★ Laplace approximation
 - ★ Adaptive Gaussian quadrature
- ▶ Maximization requires optimization algorithms
 - ★ Expectation-maximization algorithm
 - ★ Newton-Raphson algorithm
- Likelihood **inference** facilitates use of likelihood ratio test;
Wald tests are also available for regression parameters

jointModel command

```
?jointModel
```

```
?jointModelObject
```

```
ml4 <- lme(log(serBilir) ~ year + drug:year, random = ~ year | id,  
           data=pb2)
```

```
ms1 <- coxph(Surv(years, status=='dead') ~ drug + sex + I(age-50),  
            data=pb2.id, x=TRUE)
```

```
mj1 <- jointModel(lmeObject=ml4, survObject=ms1, timeVar='year',  
                 method='weibull-PH-aGH')
```

jointModel arguments

- **lmeObject**
 - ▶ Linear mixed-effects model for longitudinal marker
 - ▶ Multi-record data, ordered in 'long' format (pbc2)
 - ▶ No additional correlation structure beyond random effects
- **survObject**
 - ▶ Standard Cox regression model for censored survival outcome
 - ▶ Single-record data in same order as for lmeObject (pbc2.id)
 - ▶ x=TRUE so that design matrix is returned
- **timeVar**
 - ▶ Time variable in lmeObject
 - ▶ lmeObject and survObject must have same time scale

jointModel arguments

- **method**: Specifies the baseline hazard function, parameterization of relative risk model, and procedure for numerical integration
 - ▶ weibull-PH-aGH (default)
 - ▶ weibull-PH-GH
 - ▶ weibull-AFT-aGH
 - ▶ weibull-AFT-GH
 - ▶ piecewise-PH-aGH
 - ▶ piecewise-PH-GH
 - ▶ spline-PH-aGH (allows strata)
 - ▶ spline-PH-GH (allows strata)
 - ▶ Cox-PH-aGH
 - ▶ Cox-PH-GH

PH: proportional hazards; AFT: accelerated failure time

GH or aGH: standard or adaptive Gauss-Hermite quadrature

Results summary

- Coefficients (SEs) from mixed-effects model and joint model

Variable	Mixed model	Joint model
Year	0.176 (0.018)	0.182 (0.018)
Year-by-treatment	0.003 (0.024)	0.005 (0.024)

- Coefficients (SEs) from extended Cox model and joint model

Variable	Cox model	Joint model
Treatment	−0.026 (0.173)	−0.034 (0.185)
Sex	0.180 (0.235)	0.163 (0.250)
Age	0.068 (0.009)	0.065 (0.009)
Serum bilirubin	1.465 (0.094)	1.358 (0.101)

Model diagnostics

Evaluate adequacy of the fitted joint model using **residual analysis**

- **Longitudinal submodel**

- ▶ Subject-specific (or, conditional) residuals

$$r_i(t) = y_i(t) - \{x_i(t)\hat{\beta} - z_i(t)\hat{b}_i\}$$

with focus on constant variance and normality

- ▶ Marginal (or, population-averaged) residuals

$$r_i = y_i - X_i\hat{\beta}$$

with focus on specification of the mean model

- **Survival submodel:** Martingale residuals

$$r_i(t) = N_i(t) - \int_0^t \hat{h}_0(s) R_i(s) \exp\{w_i \hat{\gamma} + \hat{\alpha} \hat{m}_i(s)\} ds$$

with focus on specification of the model for $m_i(t)$

Advanced models

Facilitate flexible modeling to fully utilize longitudinal information

- **Interaction effects** (§5.1.1)

$$h_i(t) = h_0(t) \exp\{w_i\gamma + [w_{i1} \times m_i(t)]\alpha\}$$

Example: Interaction between treatment and bilirubin on mortality

- **Time-dependent slopes** (§5.1.3)

$$h_i(t) = h_0(t) \exp\{w_i\gamma + \alpha_1 m_i(t) + \alpha_2 m'_i(t)\}$$

Example: Effect of level and slope of serum bilirubin on mortality

Advanced topics

- Discrete markers (§5.7)
- Competing risks (§5.5.1)
- Recurrent events (§5.5.2)
- Mediation (Richiardi et al., 2013)
- Prediction (§7)

Analysis goals

- **Estimation**

- ▶ Characterize association between exposure and outcome
- ▶ Control for observed confounders, modeling effect modification
- ▶ What is the association between serum bilirubin and mortality?

- **Prediction**

- ▶ Predict average outcome from covariate values
- ▶ Model selection focuses on maximizing prediction accuracy
- ▶ Can serum bilirubin be used to predict mortality risk?

(Shmueli, 2010; French et al., 2016)

Dynamic prediction

Update predictions over time based on new longitudinal data

- Prediction of conditional survival probability

$$\pi_i(u | t) = P[T_i^* \geq u | T_i^* > t, \mathcal{Y}_i(t), w_i, \mathcal{D}_n; \theta]$$

at time $u > t$ given survival to time t

- Prediction of longitudinal outcome

$$\omega_i(u | t) = E[Y_i(u) | T_i^* > t, \mathcal{Y}_i(t), w_i, \mathcal{D}_n; \theta]$$

at time $u > t$ given observed responses $\mathcal{Y}_i(t)$

given sample data $\mathcal{D}_n = \{T_i^*, \delta_i, y_i; i = 1, \dots, n\}$

Summary

- Longitudinal model + Survival model = Joint model
 - ▶ Describe evolution of marker levels over time for each subject
 - ▶ Associate the subject-specific evolutions with event hazard
 - ▶ Specify a dependence structure between the two models
- Combine a linear mixed-effects model for longitudinal marker with a relative risk model for a censored survival outcome
- Advantages over separate longitudinal and survival analyses
 - ▶ Incorporates random error term for measurement error
 - ▶ Accommodate marker as an endogenous time-dependent covariate
 - ▶ Marker is not assumed constant in between measurement times
 - ▶ Flexible modeling to fully utilize longitudinal information
- Model fitting can be complex and computationally intensive;
analysis methods should be selected based on the scientific question
- Software: R, JM::jointmodel

Textbooks

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2. Rizopoulos D. Joint Models for Longitudinal and Time-to-Event Data with Applications in R, 2012.
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Resources

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